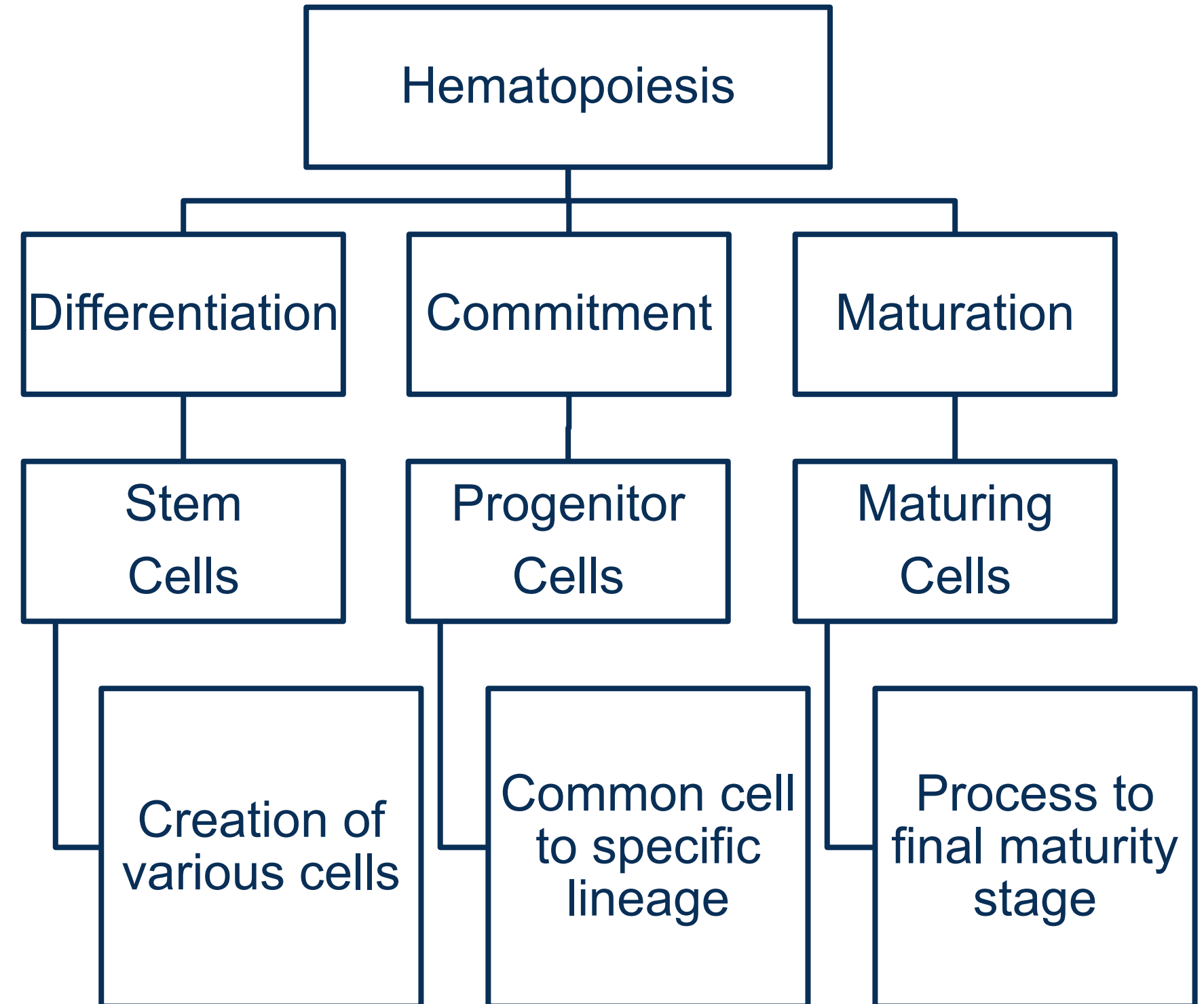


Hematopoiesis

Hematopoiesis

- Maintain Tissue Homeostasis
 - Maintenance of adequate number of cells for normal body functions
 - Balance between cell proliferation, differentiation, apoptosis
- Governed by cytokines

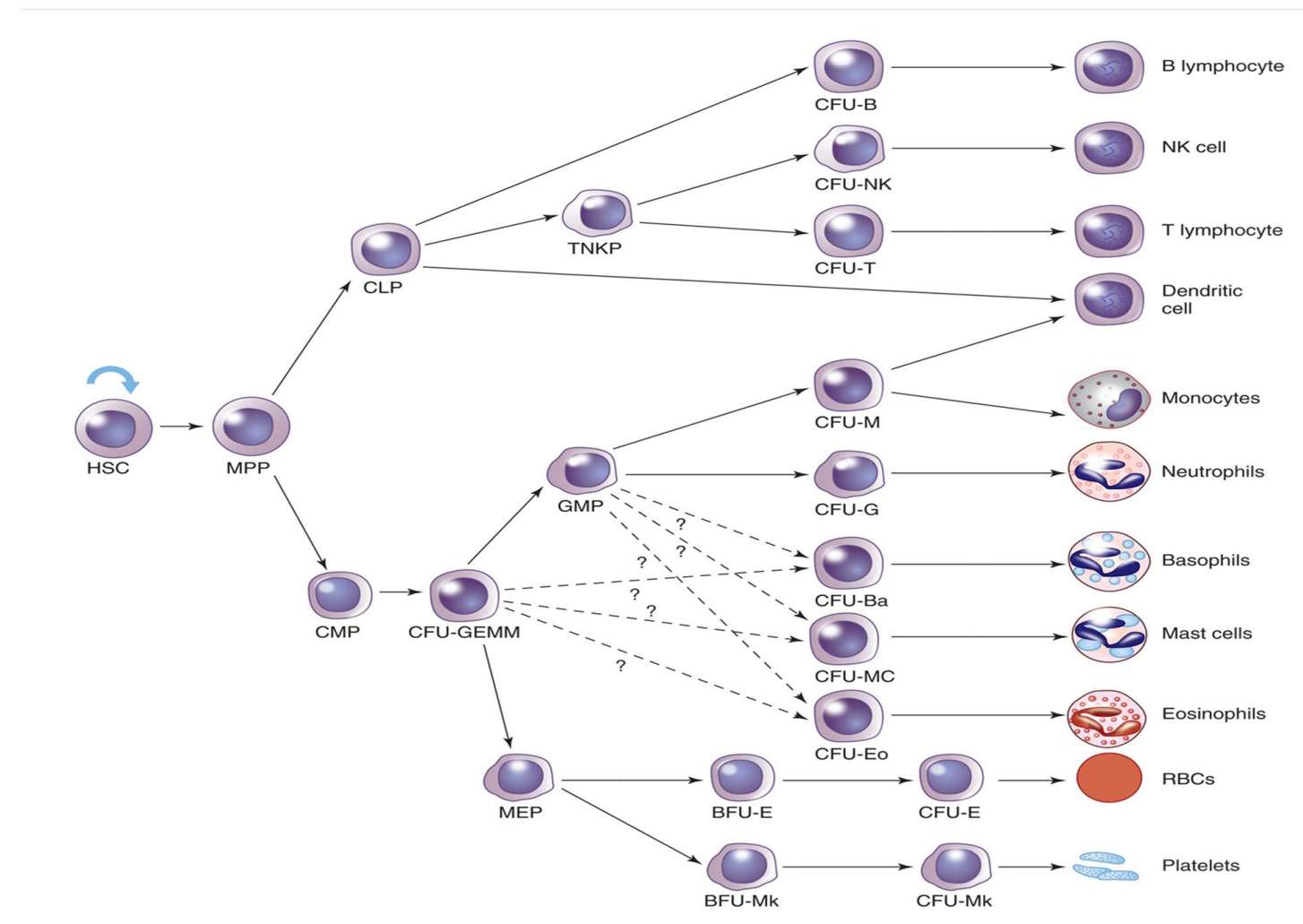


Hematopoiesis

Bone Marrow

Hematopoietic precursor cells

- Hierarchical structure of precursors to mature cells
- Stem cells > progenitor cells > maturing cells

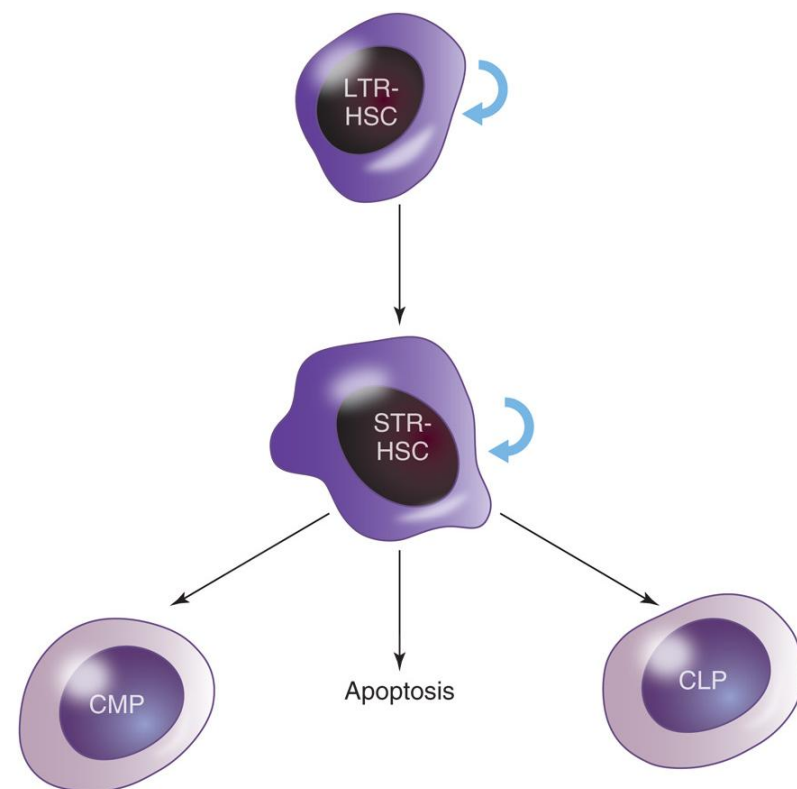


Peripheral Blood

Peripheral blood cells

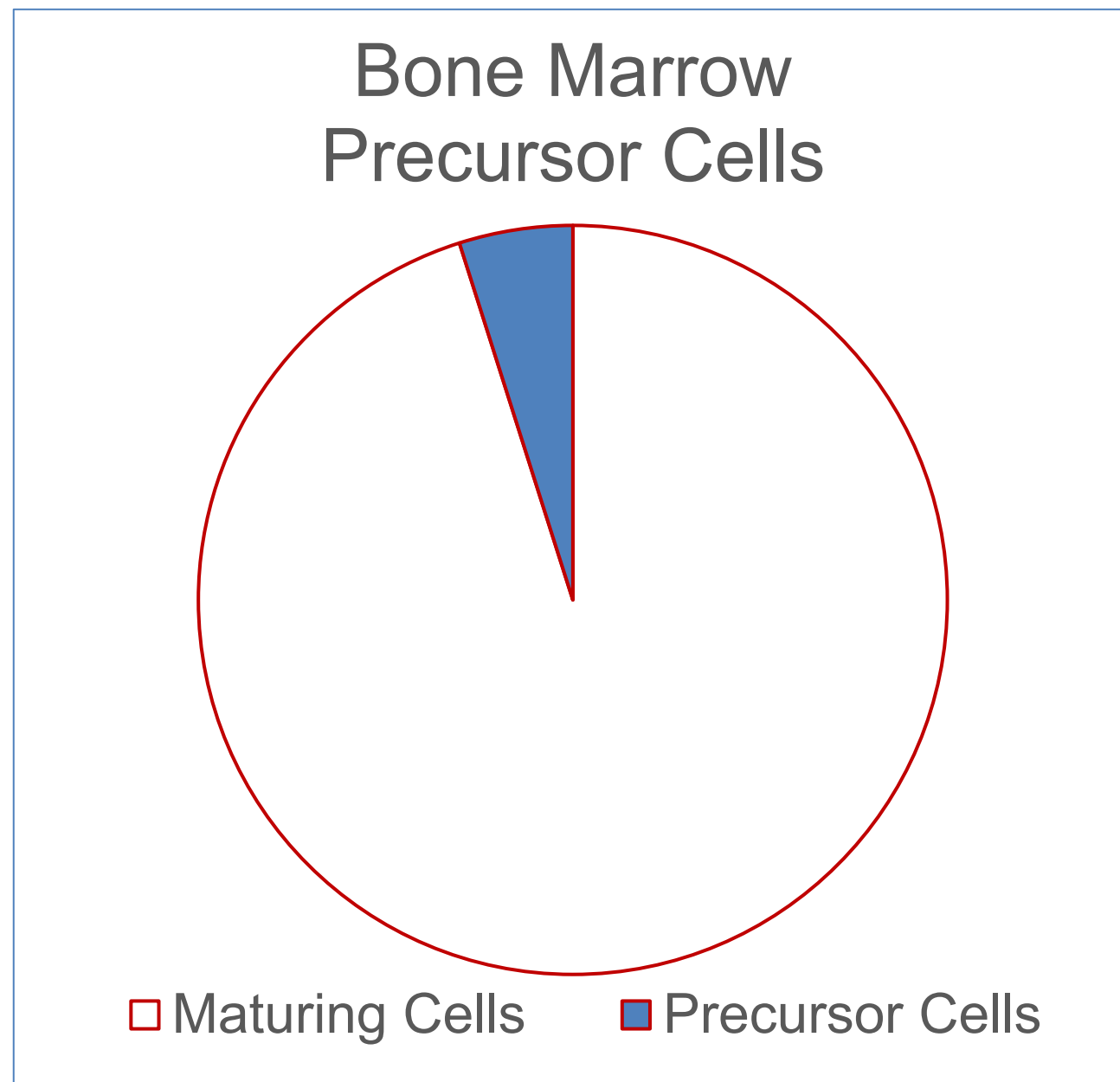
- Mature cells in healthy individual
- Incapable of mitosis
 - *except lymphocytes
- Terminally differentiated

Model for Precursor Cells

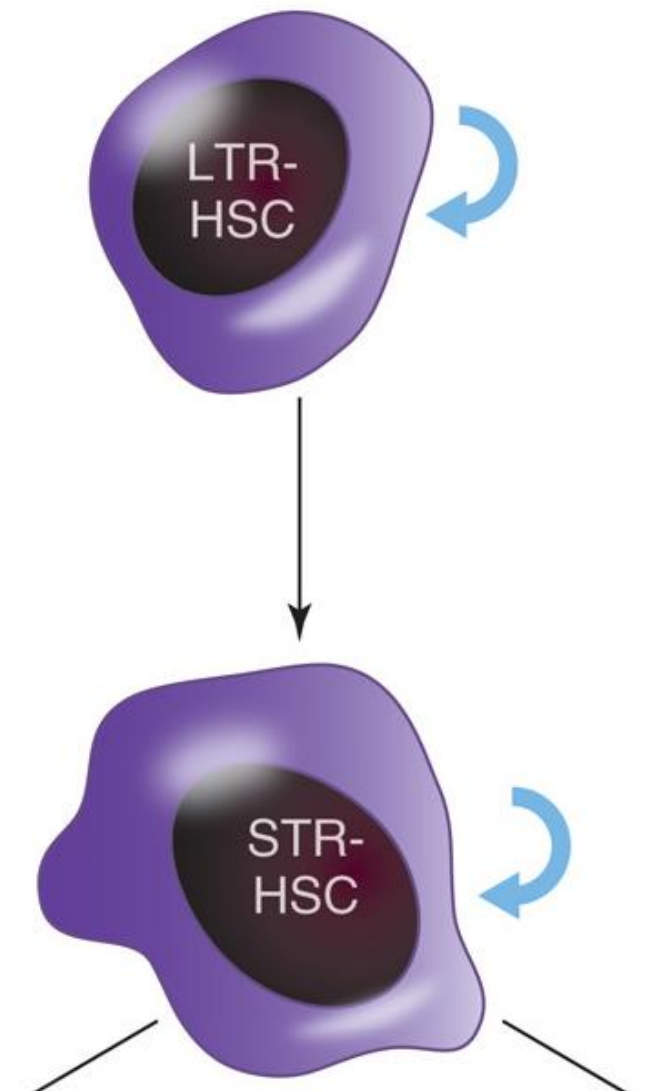


- Pluripotential HSC (LTR and STR)
 - Self renews
 - Gives rise to all hematopoietic elements
- Earliest differentiating cells
 - CMP = common myeloid progenitor
 - Neutrophils, monocytes, eosinophils, basophils, erythrocytes, megakaryocytes
 - CLP = common lymphoid progenitor
 - T and B cells, natural killer cells, lymphoid dendritic cells

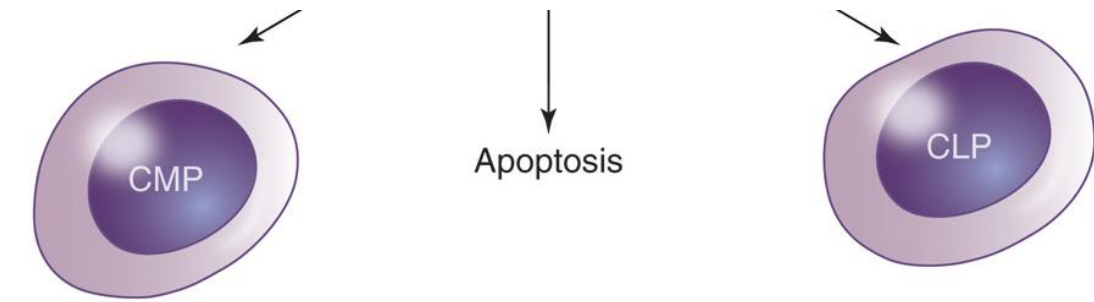
Stem Cells



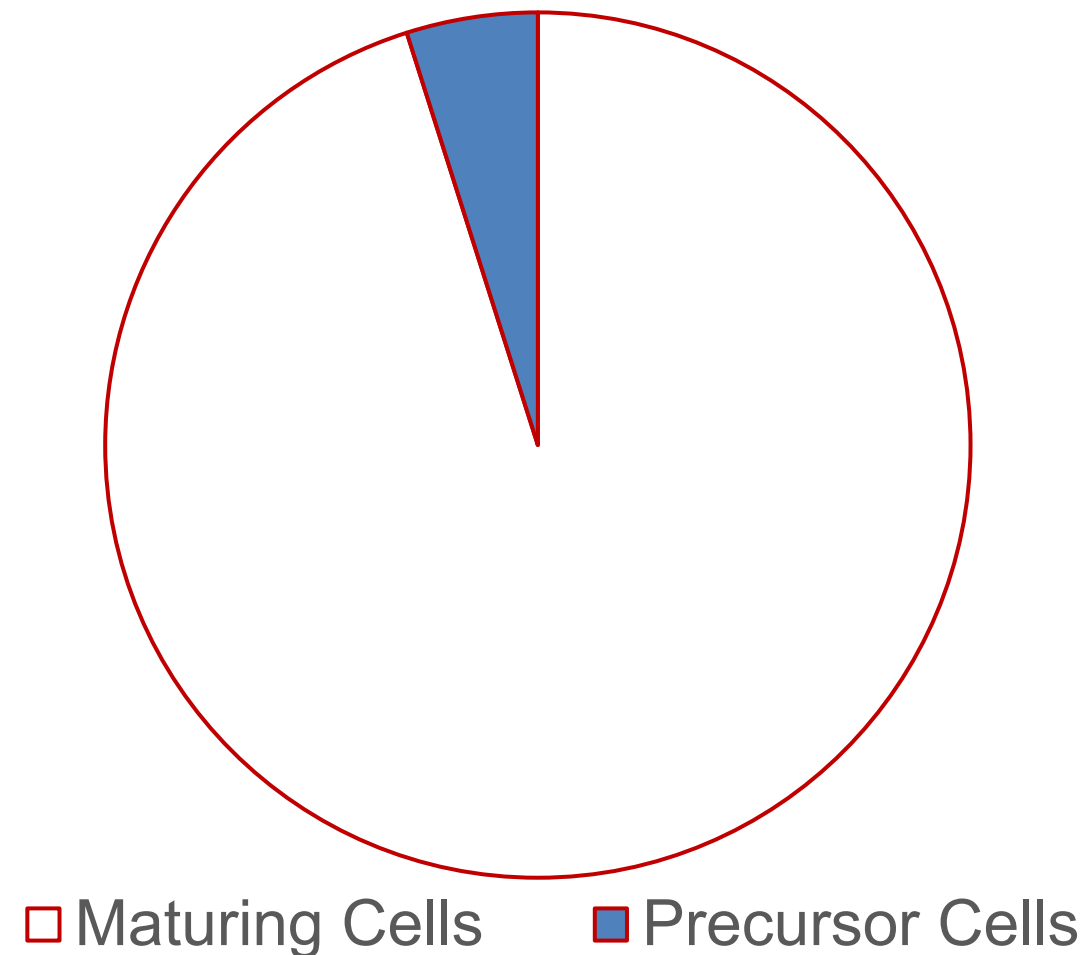
- Undifferentiated cells (HSCs)
- Asymmetric cell division
- Four possible fates
 - Quiescence
 - Self-renewal
 - Commitment via CMP or CLP
 - Apoptosis
- Cell fate dependent on cellular environment (BM niches)



Progenitor Cells



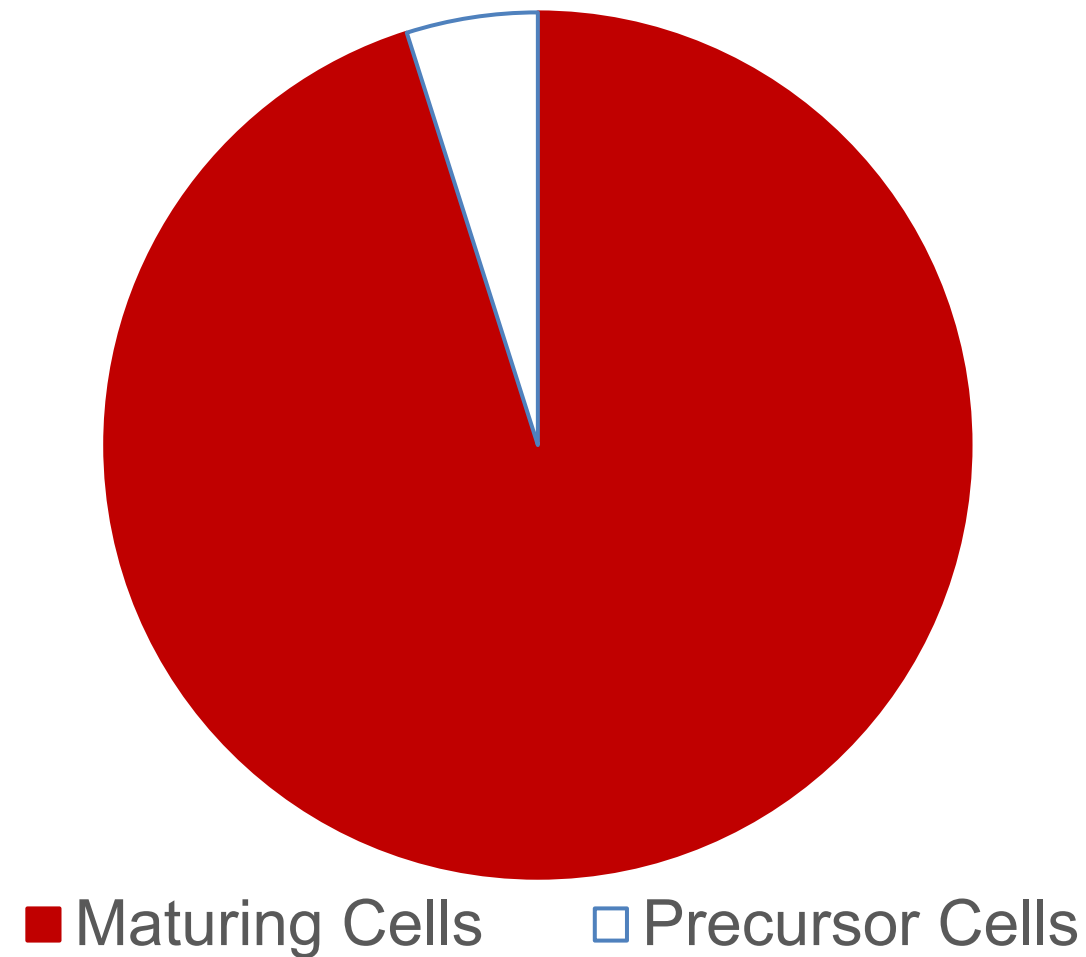
Bone Marrow
Precursor Cells



- Originate from stem cells as multipotential progenitor cells (MPPs)
 - Downregulation of HSC genes
 - Upregulation of lineage specific genes
- Cells transition to hematopoietic progenitor cell compartment (HPC)
- Mitotically more active than HSCs
- Gradually becomes restricted in differential potential

Maturing Cells

Bone Marrow
Maturing Cells



- >95% of hematopoietic precursor pool
 - Committed cell population
- Morphologically recognizable by
 - Nuclear and cytoplasmic characteristics
 - Earliest cell “Blast”
 - Lymphoblast, myeloblast, pronormoblast, megakaryoblast

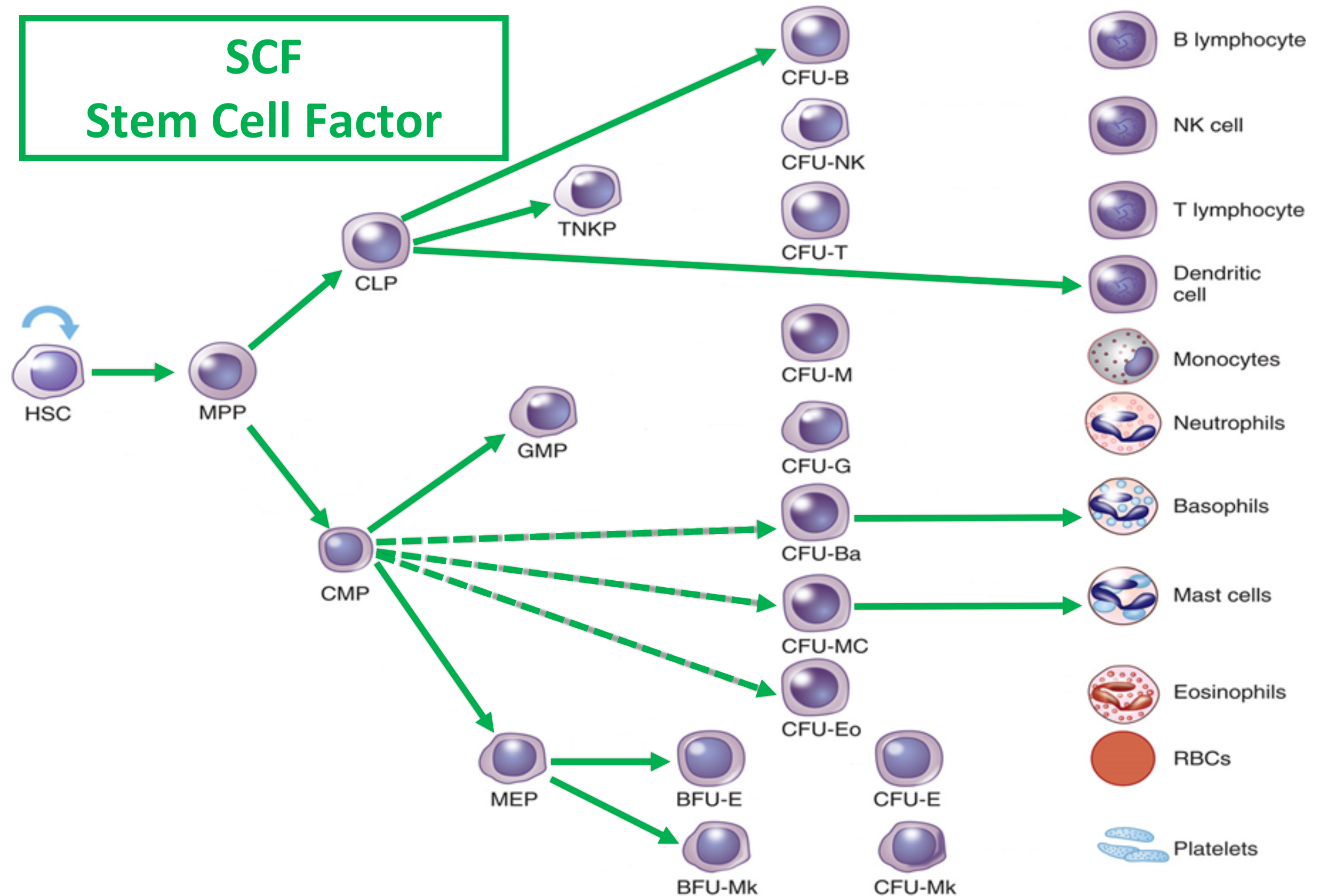
Cytokines (growth factors)

- Cell to cell communication to govern precursor cell survival, self-renewal, proliferation, differentiation
 - Historically = CSFs or **colony-stimulating factors**
 - New nomenclature = Interleukins
- Most produced by stromal cells
 - *except erythropoietin (EPO), kidney
- Most influence similar functions (redundancy) and multiple biologic activities (pleiotrophy)

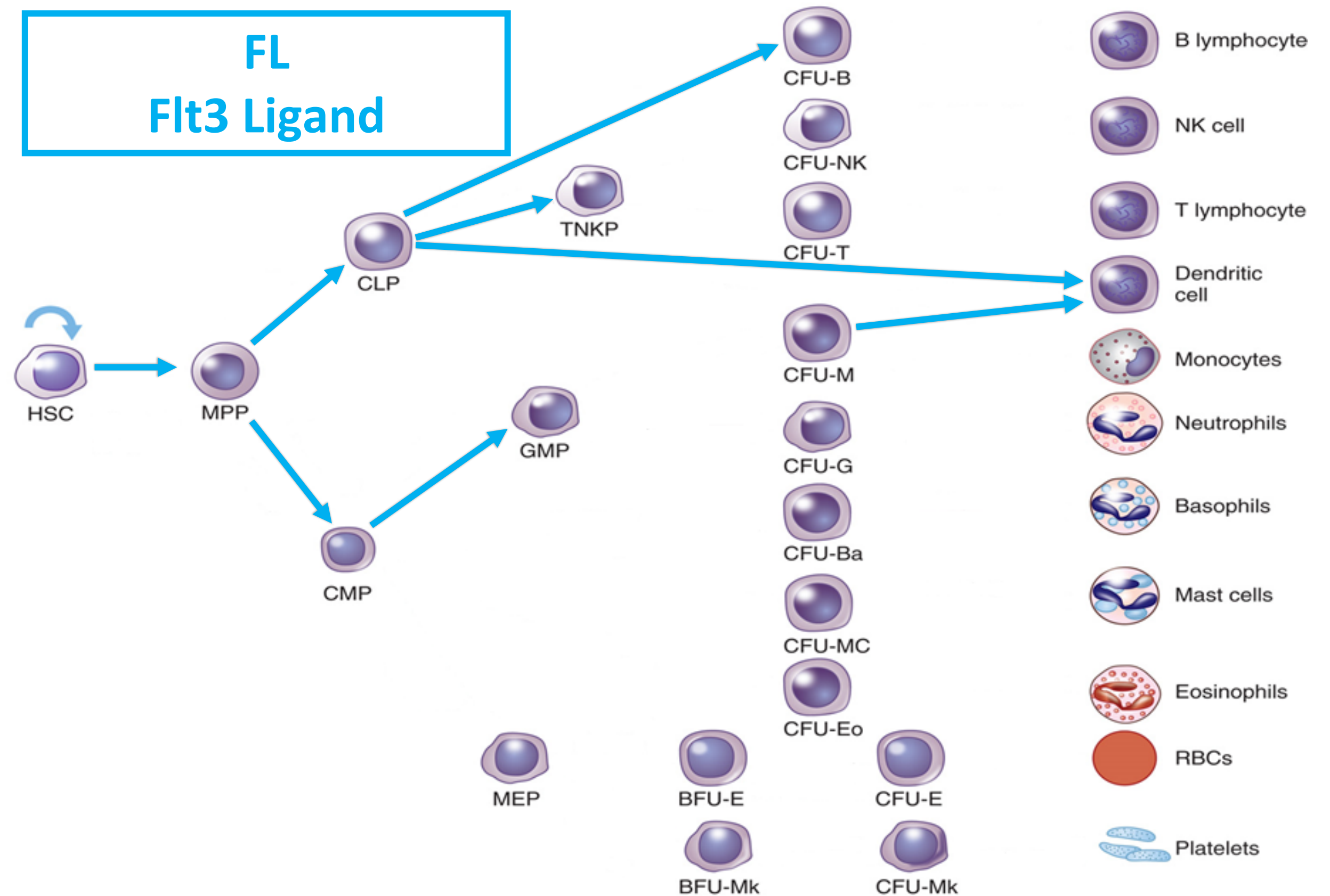
Cytokine Promotors

- Early-acting (multilineage)
 - Effect on MPP cells
 - Affects proliferation and differentiation
 - SCF, FL, GM-CSF, IL-3, IL-6, IL-11, others
- Later-acting (lineage-restricted)
 - Effect on committed cells
 - Affects maturation
 - G-CSF, M-CSF, EPO, TPO, IL-5, others

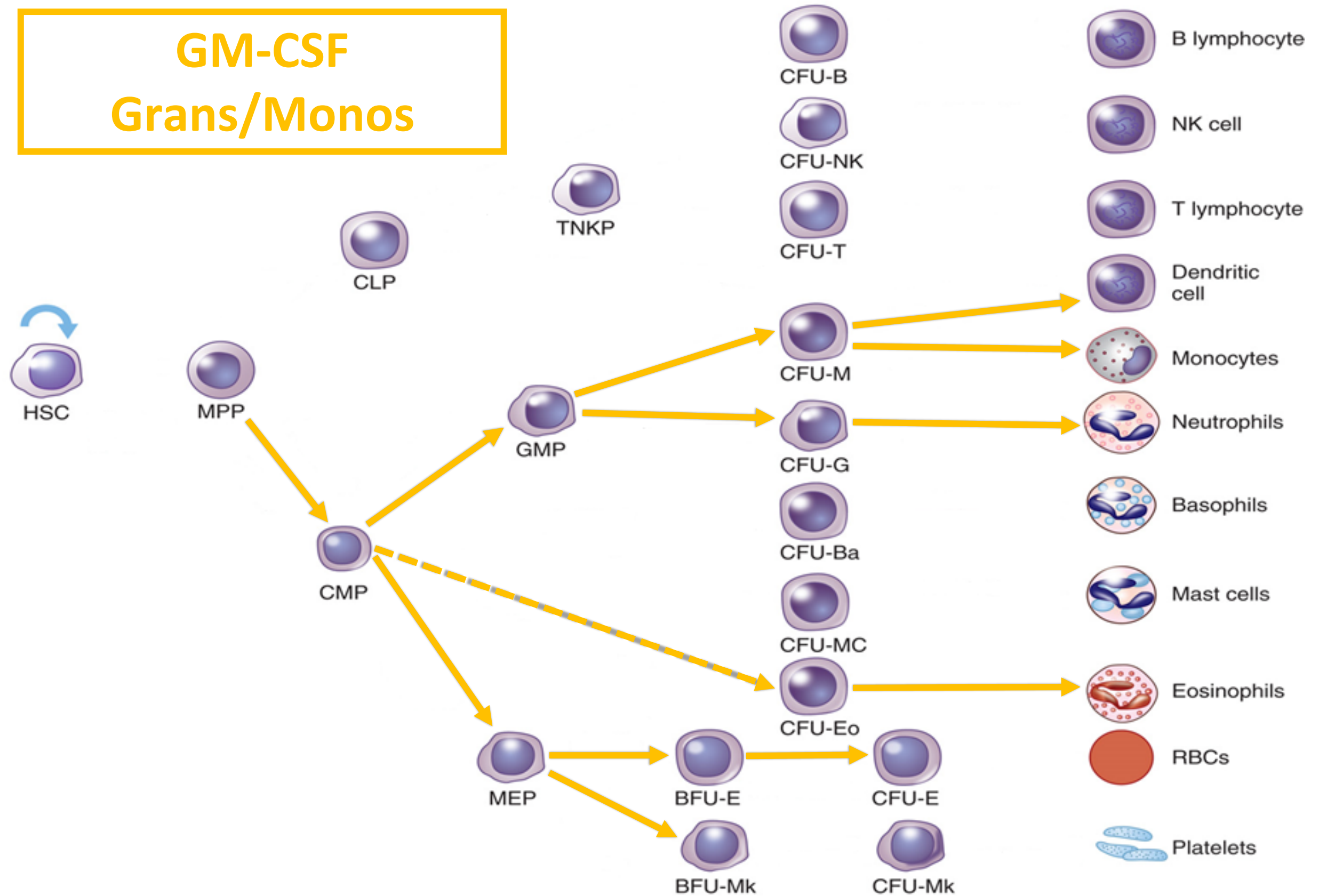
Cytokine Early Promotors



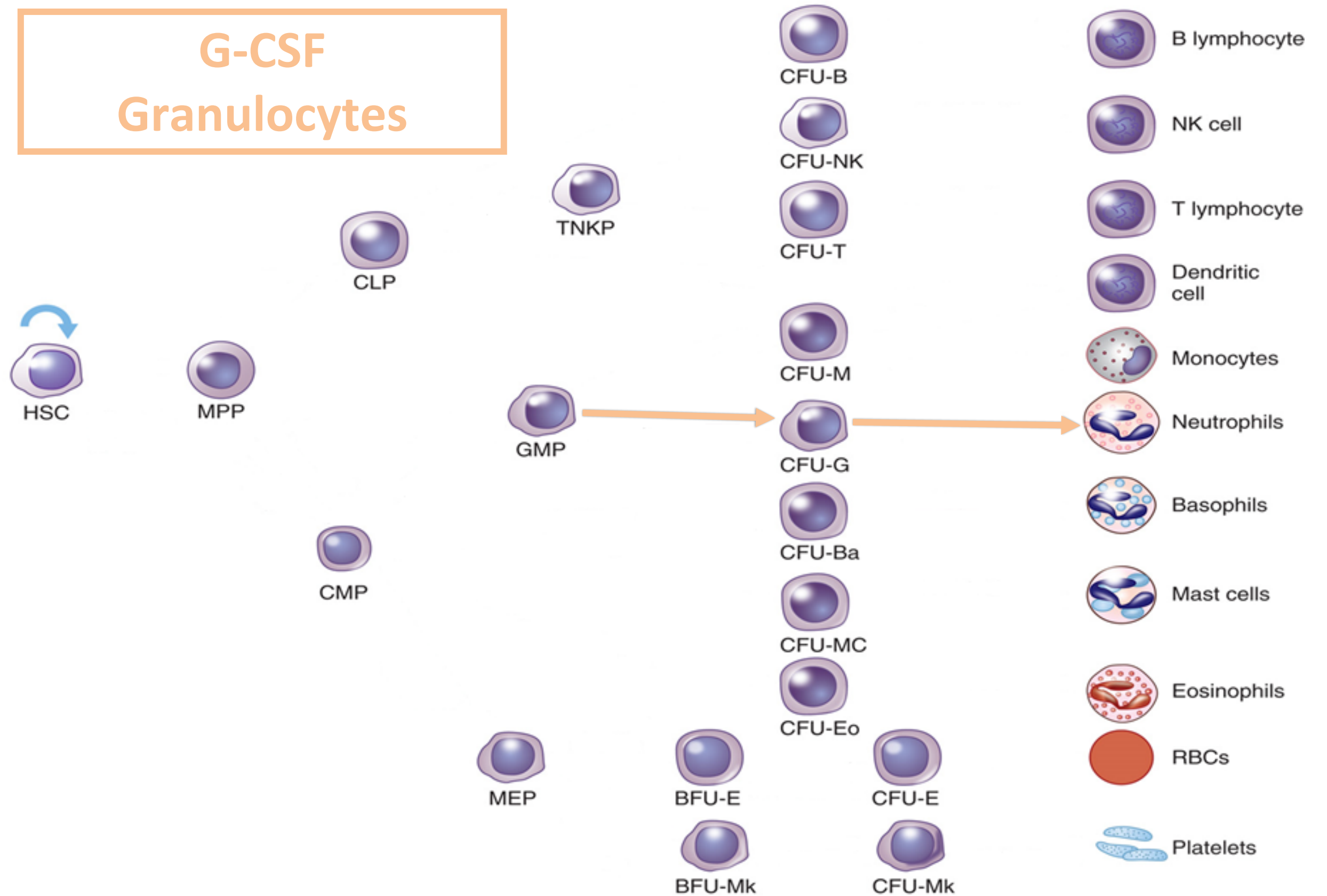
Cytokine Early Promotors



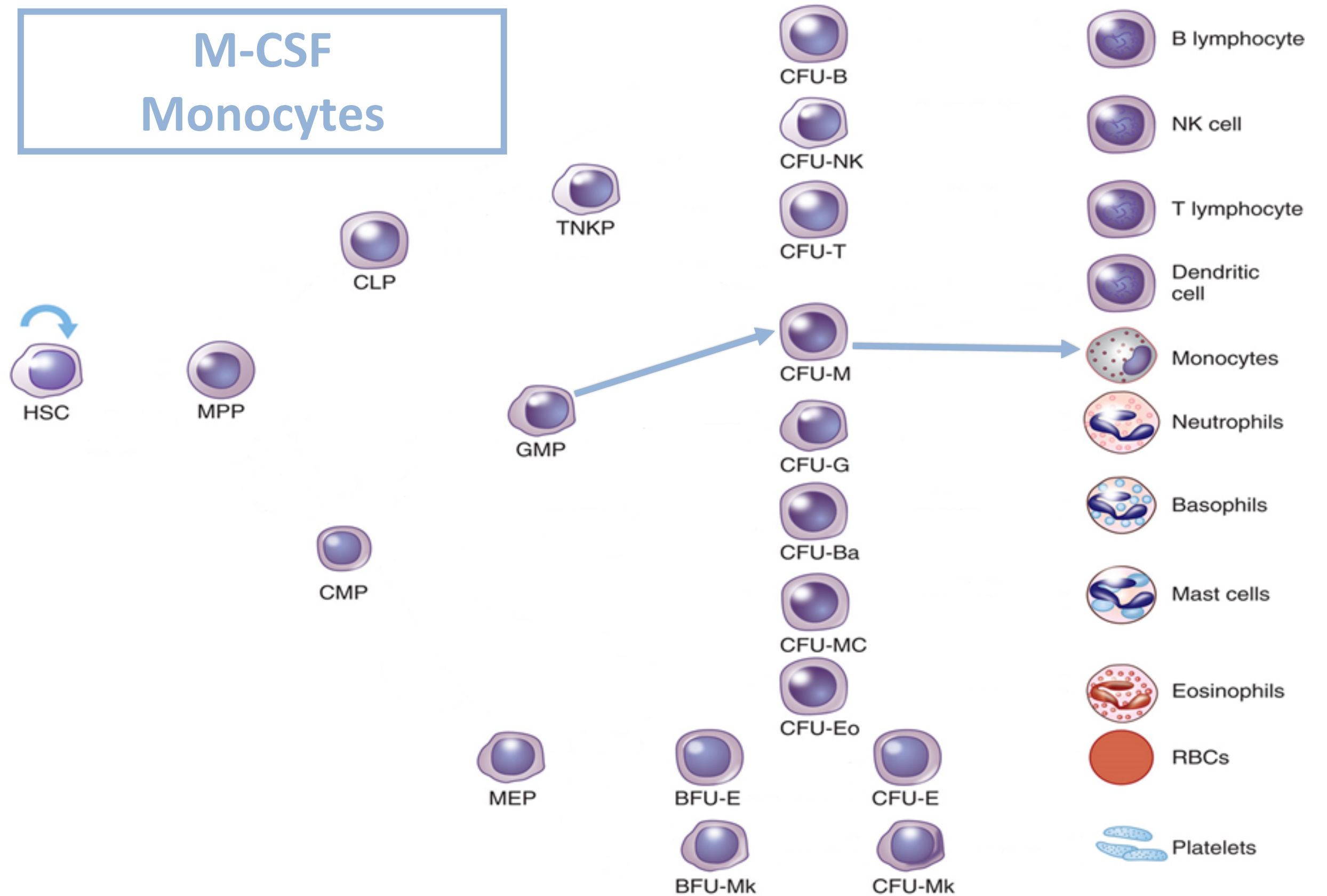
Cytokine Early Promotors



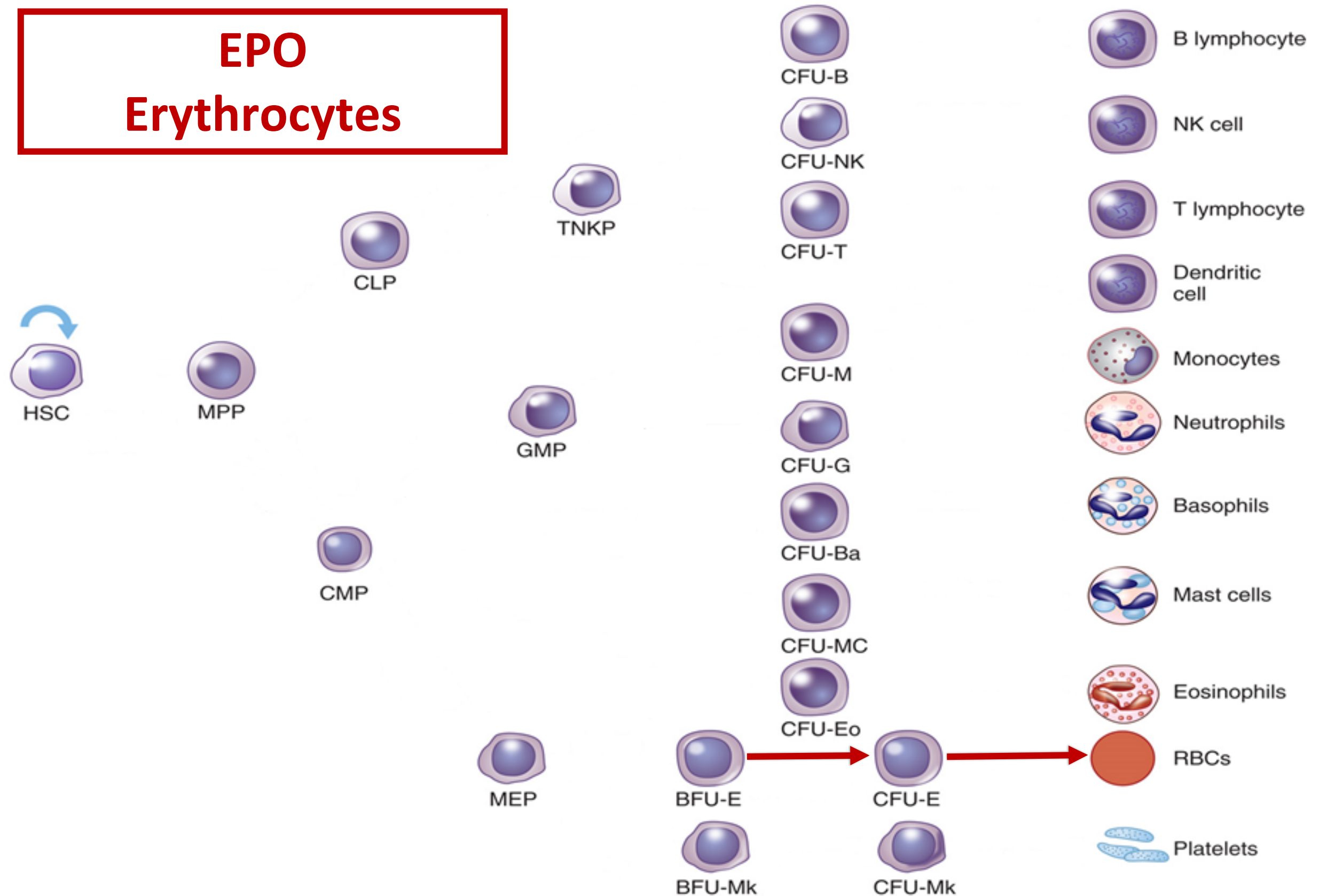
Cytokine Late Promotors



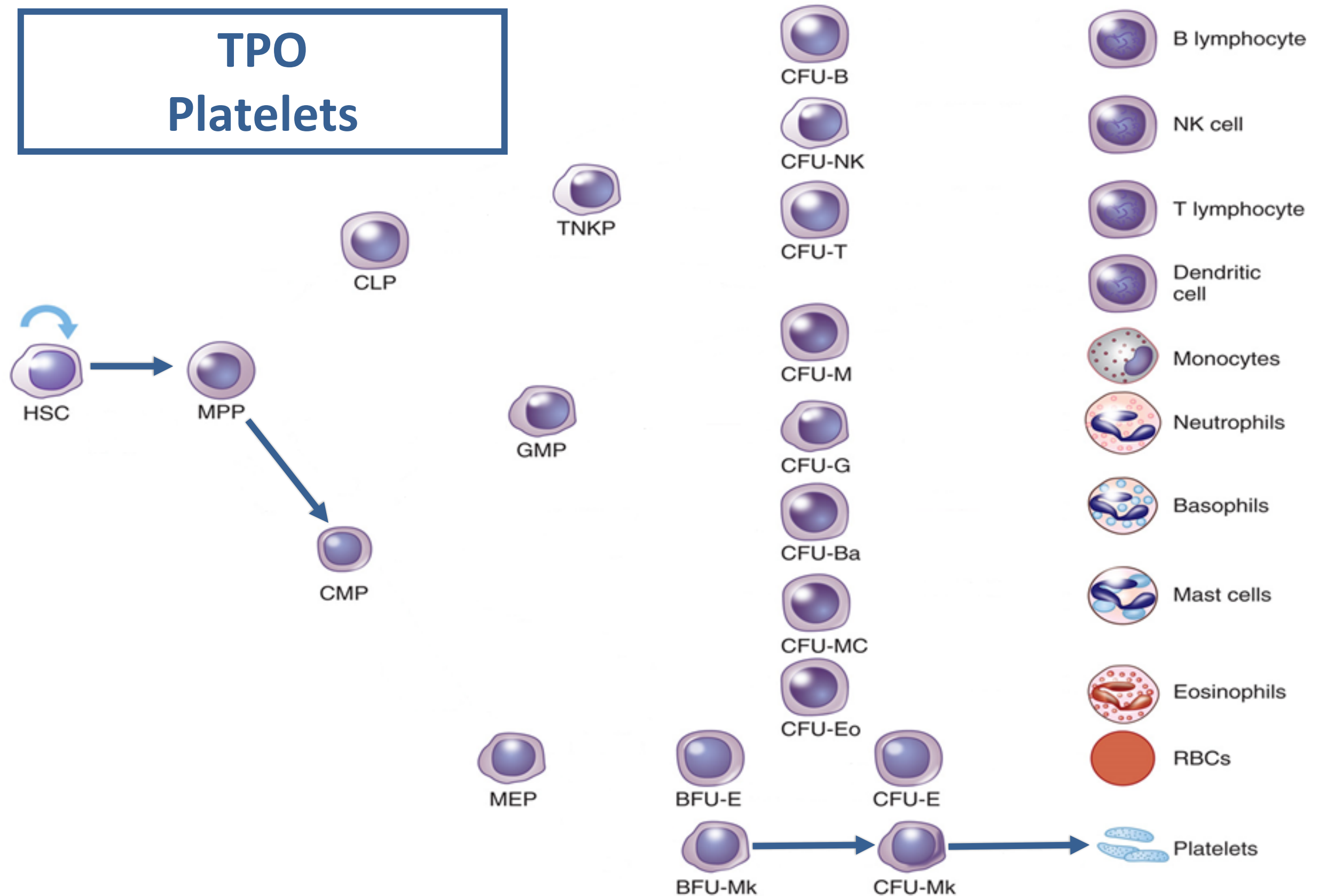
Cytokine Late Promoters



Cytokine Late Promoters



Cytokine Late Promotors

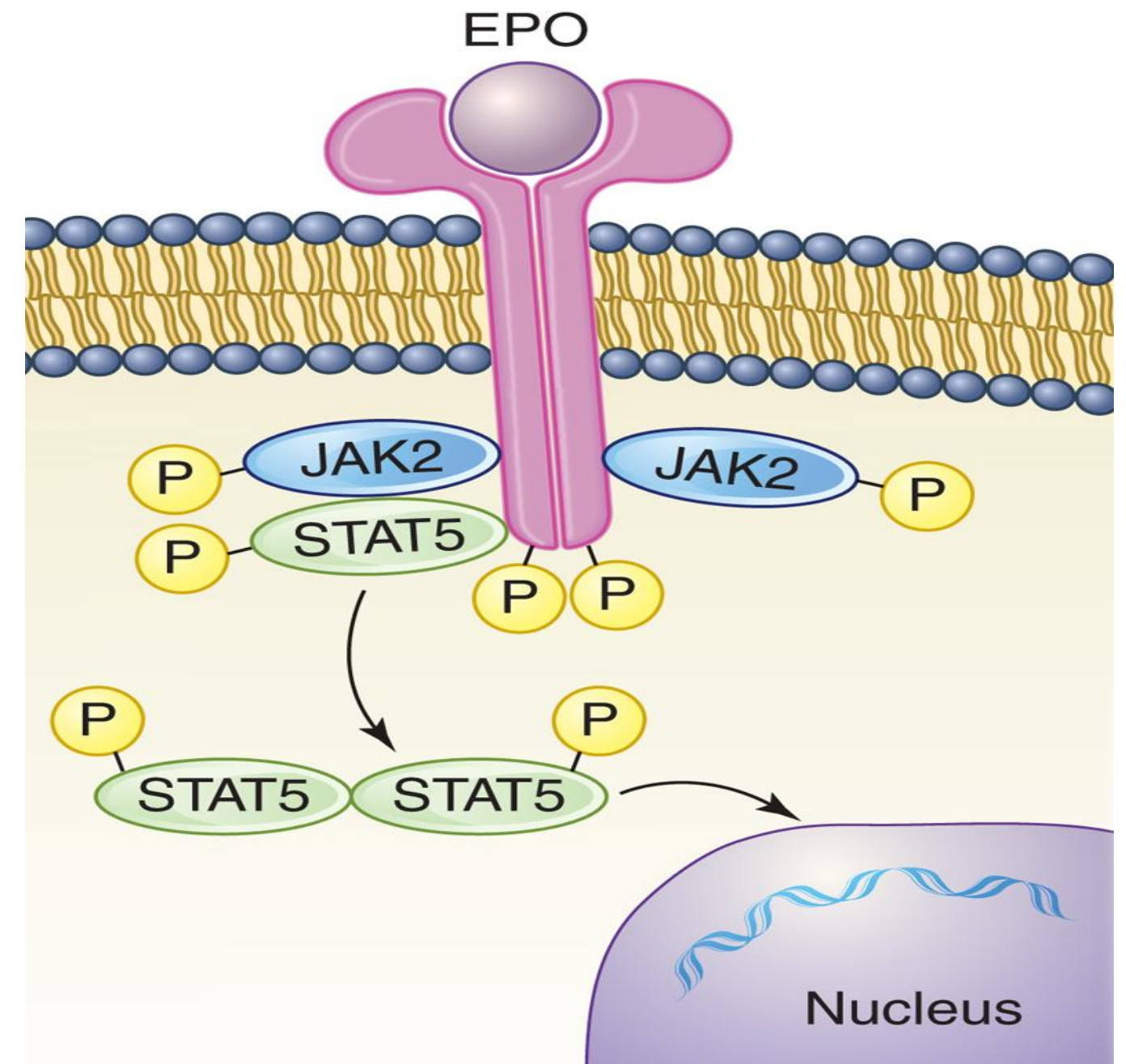


Hematopoiesis Regulation

- Cytokines inhibit precursor cell production
 - Decrease production of cytokine promoters
 - Increase production of cytokine regulators
- T and NK cells act

Cytokines Communication

- Most signaling pathways via protein phosphorylation
- JAK2 mutation common in myeloproliferative disorders
 - Polycythemia vera



Cytokine Clinical Use

- EPO
 - Anemias
- G-CSF/GM-CSF
 - Granulocytopenia
 - Auto/allogeneic stem cell collections
- Others

Cell Surface Proteins

Cluster of Differentiation

